Output Form (OF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: *MRBENCH* | ***Context:*** *Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

LOWER priority per elicitation. Output form (categorical labels [Q58, Q59], DAMR percentages, and Pearson correlations [Q64, Q66]) is appropriate for the deployment scenario where teachers render acceptability judgments on AI+human-augmented responses. The benchmark explicitly avoids unreliable NLG metrics like BLEU/BERTScore [Q4, Q7, Q8] in favor of pedagogically meaningful categorical scores. No mismatch with deployment output needs is identified. Minor concern: the benchmark reports per-dimension DAMR but no aggregate acceptability score weighted by deployment priorities; teams would need to compute their own weighted aggregate to map to a binary acceptable/unacceptable judgment, but this is straightforward.

ID	Evidence
Q4	“General domain-agnostic natural language generation (NLG) metrics (Lin, 2004; Popovic´, 2017; Post, 2018; Gao et al., 2020; Liu et al., 2023) are not well-suited for this context, as most of them fail to account for pedagogical values and require gold references, which are often not available, especially in online interactions.” (p.1)
Q7	“General domain-agnostic natural language generation (NLG) metrics like BLEU (Papineni et al., 2002), BERTScore (Lin, 2004), DialogRPT (Gao et al., 2020), and so on have been used as proxies to measure the coherence and human-likeness of AI tutor responses.” (p.2)
Q8	“However, these metrics do not account for pedagogical values (Jurenka et al., 2024; Liu et al., 2024) and often require a ground truth answer to evaluate matching responses.” (p.2)
Q58	“Each dimension was annotated using a three-tier labeling system (see Figure 1 and Table 4).” (p.6)
Q59	“For instance, the 'mistake identification' dimension employed the following labels: (i) yes, (ii) to some extent, and (iii) no.” (p.6)
Q64	“We utilize two key metrics to quantitatively assess the pedagogical effectiveness of LLMs and for comparative analysis: (1) Desired Annotation Match Rate (DAMR): This metric quantifies the percentage of responses from each human or LLM-based tutor that received the desired annotation labels.” (p.6)
Q66	“(2) Annotation Correlation (AC): This metric is based on Pearson's correlation (Sedgwick, 2012), and it estimates the correlation between LLM-generated and human annotations (Kim et al., 2024), allowing us to assess the reliability of LLMs as evaluators in the context of student mistake remediation.” (p.6)

Strengths

- Categorical and percentage outputs map cleanly onto teachers' acceptability judgments
- Per-dimension reporting [Q67] supports custom re-weighting toward 'providing guidance'
- Explicit rejection of unreliable surface-level NLG metrics [Q4, Q7, Q8]
- Both DAMR (per-tutor performance) and AC (LLM-evaluator reliability) metrics provided [Q64, Q66]

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Q67	“Table 3 shows DAMR scores for each LLM across all eight dimensions.” (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (categorical labels and percentages) matches the deployment, where teachers judge acceptability of text responses [Q58, Q64]. No mismatch.

OF-2: Text-to-speech is not relevant — both benchmark and deployment are text-only.

OF-3: Target population is fluent in English with high literacy and professional qualifications [WEB-1, WEB-2]; accessibility is not a concern for the evaluator role.

OF-4: No form mismatches harming external validity. Categorical/percentage outputs are appropriate. Aggregating per-dimension DAMR into a deployment-relevant acceptability score requires user-side weighting but is methodologically simple.

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Q133	“The prompt template used to generate responses from the seven considered LLMs for both the Bridge and MathDial datasets is shown in Figure 2.” (p.12)
WEB-1	Indian Grade 1–8 teachers in CBSE schools require B.Ed/D.El.Ed and pass CTET — high literacy and professional qualification (https://www.asiancollegeofteachers.com/blogs/1524-Teaching-License-In-India-What-It-Is-And-How-Do-You-Renew-It-blog.php)
WEB-2	CTET eligibility confirms English literacy among target population (https://testbook.com/ctet/eligibility-criteria)